

Soybean Seeds Quality Identification through Image Classification Using Machine Learning

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Abstract—Soybean Seeds quality is essentially important for the crop production since seed defects are one of the main factors affecting the crop yield and its economic value. Traditional methods of manual inspection are not only time, consuming. To tackle the problem, the authors of this work have proposed an automated image, based soybean seeds quality classification framework through deep learning models.

In this research manually collected healthy soybean seeds class dataset and five defective seeds classes have been obtained from publicly available Kaggle repositories. The dataset was subjected to several standard preprocessing operations such as image resizing, normalization, and data augmentation to improve robustness and generalization capability. The custom CNN model, several deep learning models were implemented and evaluated, including pre, trained architectures such as VGG16, MobileNetV2, and ResNet50 using transfer learning. Model performance was measured with the help of a set of evaluation metrics that are well, known and widely used, such as accuracy, precision, recall, F1, score, and confusion matrix analysis. Experiments show that the VGG16 model achieved the highest classification accuracy of 85%.

Index Terms—Soybean Seed Classification, Image Processing, Seed Quality Assessment, Transfer Learning, Multi-class classification

I. INTRODUCTION

Among agricultural crops, soybean holds a remarkable place owing to its high nutritional value and diverse utilization in food, feed, and industrial applications [16]. The quality of soybean seeds is essentially a determining factor for the crop [9] yield [4] [7] and the market price of the product. In the past, the quality assessment of soybean seeds was entirely dependent on the manual inspection of experts which is a very slow process, subjective and prone to mistakes of human, especially in the case of a large volume of seeds.

Automated image, based seed classification systems have become a very effective alternative to traditional methods with the recent development of computer vision and deep learning. Deep learning models have demonstrated great ability to learn complex visual features directly from images, such as texture, shape, and surface defects. Nevertheless, the precise identification of soybean seeds [8] is still a daunting task mainly because of high inter, class similarity among defective

categories and the changes in lighting, orientation, and seeds appearance.

Here, a six, category soybean seeds quality classification [1] system based on deep learning techniques is proposed. The dataset includes one category of healthy soybean seeds [18] that was collected and curated manually by the authors, and five unhealthy categories which are different types of defects collected from the publicly available Kaggle datasets. In order to achieve a more consistent learning effect, the dataset was prepared by resizing, normalization, and data augmentation methods.

Multiple deep learning models [2] [3], including a custom CNN, VGG16, MobileNetV2, and ResNet50 [15], were implemented and evaluated. The models were compared using standard performance metrics such as accuracy, precision, recall, F1-score, along with confusion matrix analysis. The primary objective of this work is to identify an effective and practical deep learning model [17] for automated soybean seeds quality classification.

II. BACKGROUND

Recent studies demonstrate that deep learning has significantly improved seed quality classification by learning discriminative visual features automatically. A detailed review of nearly 18–20 research papers highlights the effectiveness of CNN-based architectures for agricultural image analysis [18]. However, variations in dataset quality, model depth, and training strategies still affect classification performance. Most existing works focus on limited models or small experimental comparisons. Motivated by these gaps, this research systematically evaluates multiple deep learning models on soybean seed images. The study aims to provide a clearer performance comparison to support reliable and automated soybean seed quality assessment.

III. METHODOLOGY OF RESEARCH WORK

This Methodology discusses the total method implemented in this study to identify different quality classes of soybean seeds by deep learning techniques [15].

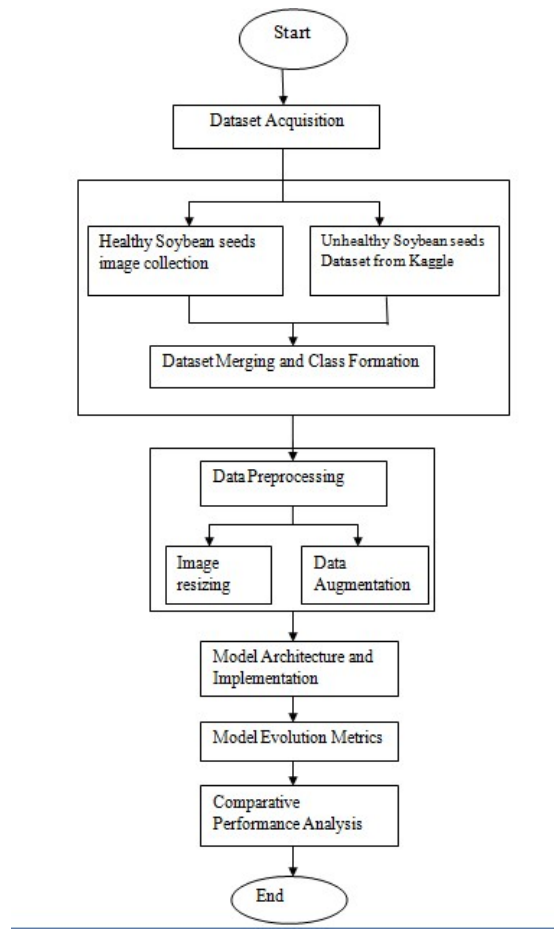


Fig. 1. Flow Diagram of Methodology

A. Healthy Soybean Seeds Image Collection

The healthy Soybean seeds [19] [20] images were directly collected by the researcher to maintain dataset originality and quality. The images were captured by manually positioning the soybean seeds in a controlled environment to reduce uneven lighting and background noise [13].

B. Unhealthy Soybean Seeds Dataset from Kaggle

[1] Images showing the different types of defects of unhealthy soybean seeds were taken from publicly available datasets on Kaggle. [11]

C. Dataset Merging and Class Formation

The image datasets of healthy [10] [18] and unhealthy soybean seeds were combined into a single dataset.

D. Dataset prepration

Data preprocessing was applied to improve image quality and ensure uniform input to the deep learning models.

1) *Image Resizing and Normalization*: All soybean seed images were resized to 224×224 pixels to maintain consistency and ensure compatibility with deep learning and pre-trained models. [5]

2) *Data Augmentation*: In order to prevent the model from overfitting and to balance the dataset, the researcher carried out data augmentation on the training images [14]. In an attempt to generate natural data variations, they employed a few random changes such as rotations, flips both horizontally and vertically, zooming, and brightness, contrast adjustment.

The model effectively gets to see the same soybean seed in a different aspect through these augmented images acting as new training samples.

3) *Dataset Splitting*: After preprocessing, the dataset was thoughtfully divided into three distinct parts for a fair and trustworthy evaluation. The training set, which contained 70% of the data, was utilized to train the model so it could learn the main features. A validation set of 15% was employed for model parameter adjustment and overfitting prevention during training. The last 15% of data was reserved as a testing set and used exclusively for overall performance evaluation.

E. Model Architecture and Implementation

To evaluate the effectiveness of different deep learning architectures [11], multiple models were implemented:

1) *Convolutional Neural Network (CNN)*: [15] A custom CNN architecture was designed as a baseline model consisting of convolutional, pooling, and fully connected layers. The CNN learns discriminative features directly from soybean seed images.

2) *Transfer Learning Models*: To improve classification performance, pre-trained deep learning [12] models were employed using transfer learning:

- VGG16 – A deep architecture with uniform convolutional layers, effective for extracting detailed visual features.
- MobileNetV2 – A lightweight and efficient model optimized for faster computation with lower resource requirements.
- ResNet50 – A deep residual network designed to address the vanishing gradient problem and enable stable training of deeper models.

The final classification layer of each pre-trained model was modified to output six classes corresponding to the soybean seed categories.

F. Model Training and Optimization

All deep learning models were trained using the same training configuration to ensure a fair and unbiased comparison.

- Categorical Cross-Entropy was used as the loss function, as it is well-suited for multi-class classification problems.
- The Adam optimizer was employed to achieve faster and stable convergence during training.
- A fixed batch size was maintained across all experiments for consistency.

The number of epochs was selected experimentally based on model performance and convergence behavior.

Model weights were updated using backpropagation, and validation loss was continuously monitored to prevent overfitting.

G. Model Evaluation Metrics

The performance of the model was evaluated with a range of quantitative metrics to ensure that the analysis was reliable and in depth. Precision and recall were calculated to assess the model's capacity to accurately detect each class, precision was aimed at minimizing false positives, and recall was aimed at minimizing false negatives, The F1 score was computed to give a harmonious measure that integrates both precision and recall.

H. Performance Analysis

We performed a comparative study of all the implemented models to see how well they perform and how feasible they are for real life use. The models were compared by overall accuracy, their performance for each soybean seeds category, training time, and computational complexity.

IV. RESULT ANALYSIS

The comprehensive analysis of the experimental results achieved through the deployed deep learning models for six, category soybean seeds classification.

A. CNN Model

Soybean seeds classification through deep learning, the CNN model was considered. The CNN model achieved an overall classification accuracy of 74%, and thus the model was able to learn basic visual features like shape and texture.

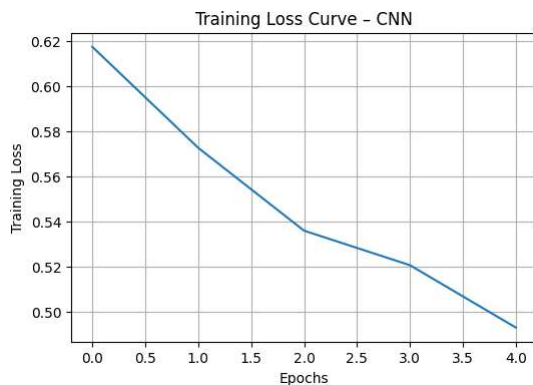


Fig. 2. Loss Graph of CNN

Training and validation accuracy graphs depict a continual increase in performance over the epochs while the loss graphs steadily lower, thus confirming the stability of the learning process.

A healthy soybean seeds was classified correctly according to the confusion matrix, but the misclassifications mainly occurred in the case of visually similar defect categories such as spotted and skin, damaged seeds. The reason for this shortcoming is the CNN's shallow architecture, which limits its capability to identify detailed defect patterns.

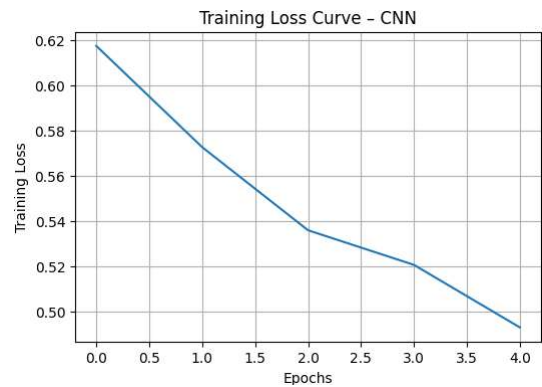


Fig. 3. Accuracy Graph of CNN

Broken Soybean	117	3	5	9	16	0
Immature Soybean	8	205	22	24	9	0
Intact Soybean	18	13	116	11	2	0
Skin-damaged Soybean	28	3	12	123	3	0
Spotted Soybean	40	10	4	4	100	0
Healthy Soybean	0	0	0	0	0	109

Fig. 4. Confusion Matrix of CNN Model

B. VGG16 Model

Compared to the baseline CNN, the VGG16 model performed better and achieved the highest accuracy of 85%. It could extract very fine visual features from pictures of soybean seeds because of its deeper and uniform convolutional structure.

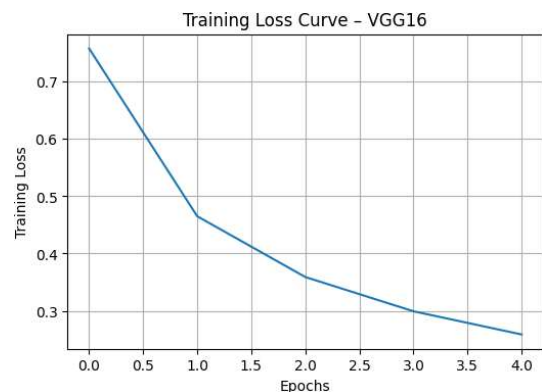


Fig. 5. Loss Graph of VGG16 Model

Training and validation accuracy plots show quite smooth convergence with very little fluctuation, whereas the loss

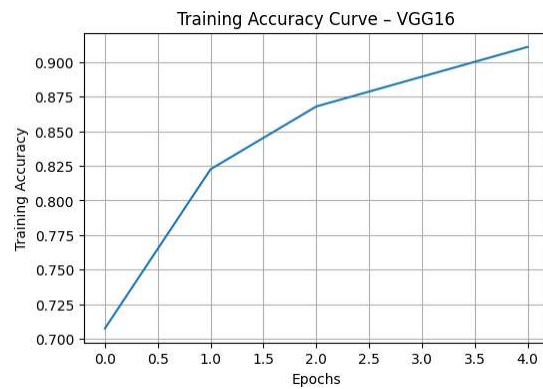


Fig. 6. Accuracy Graph of VGG16 Model

curves are continuously going down, which is an indication of high learning stability and less overfitting.

Broken Soybean	121	3	1	7	18	0
Immature Soybean	3	136	12	2	15	0
Intact Soybean	3	30	140	0	7	0
Skin-damaged Soybean	19	4	6	136	4	0
Spotted Soybean	8	4	1	0	145	0
Healthy Soybean	0	0	0	0	0	179

Fig. 7. Confusion Matrix of VGG16 Model

A strong diagonal line in the confusion matrix is evidence that the model was able to correctly predict the categories of all six classes. Defected were reduced to almost zero, in particular for complicated defect classes such as broken and immature seeds. These results confirm the ability of VGG16 to carry out a detailed classification of soybean seeds.

C. MobileNetV2

MobileNetV2 achieved an overall accuracy of 80%, offering a good balance between classification performance and computational efficiency. Despite being a lightweight model, it demonstrated competitive results across most seed categories.

The training graphs indicate faster convergence with stable accuracy and controlled loss values. Minor fluctuations in validation loss suggest efficient generalization despite reduced model complexity.

The confusion matrix shows accurate classification of healthy and severely damaged seeds, while moderate confusion is observed among subtle defect categories. Due to its low computational cost and reasonable accuracy, MobileNetV2 is well-suited for real-time and resource-constrained agricultural applications.

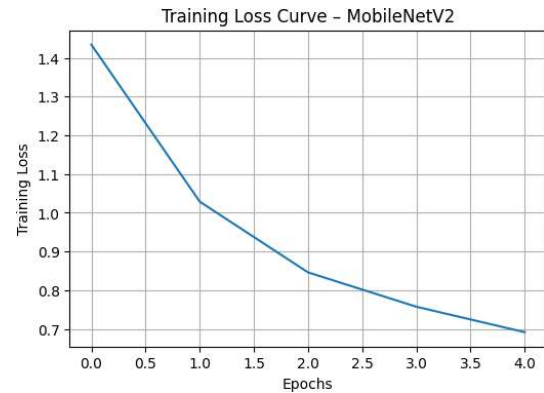


Fig. 8. Loss Graph of MobileNetV2

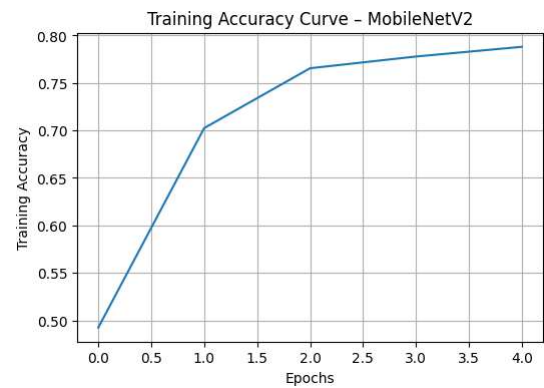


Fig. 9. Accuracy Graph of MobileNetV2

Broken Soybean	99	5	6	26	14	0
Immature Soybean	3	131	18	9	7	0
Intact Soybean	2	18	136	14	10	0
Skin-damaged Soybean	8	3	20	172	0	0
Spotted Soybean	15	6	11	1	115	0
Healthy Soybean	0	0	0	0	0	179

Fig. 10. Confusion Matrix of MobileNetV2 Model

D. ResNet50

ResNet50 scored around 80% in accuracy, which is a pretty good indication of its ability to learn well via its residual network architecture. The residual links allowed successful training of deeper layers and therefore better feature representation.

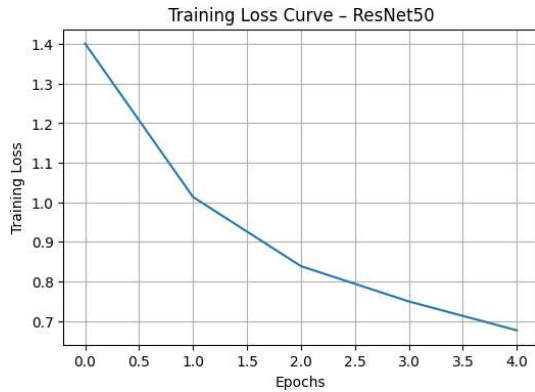


Fig. 11. Loss Graph of ResNet50

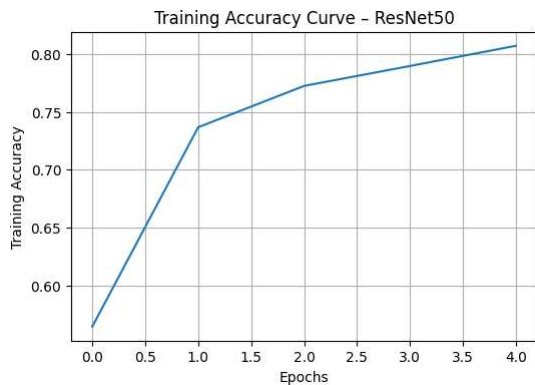


Fig. 12. Accuracy Graph of ResNet50

The figure depicting accuracy and loss over time reflects stable convergence, with the training loss declining and validation performance remaining steady. The model has been quite resilient across the different iterations and has not been heavily overfitted.

From the confusion matrix, it is clear that there has been an improvement in the class, wise discrimination when compared to the CNN model, which is especially evident in the overlapping defect categories. Even though ResNet50 demands more computational power, its strong feature extraction capability makes it an excellent choice for a precise agricultural inspection system.

The accuracy comparison shows that base CNN performed the worst with just 74% accuracy, which is likely due to an insufficient learning of features. VGG16 lead the pack at 85% accuracy since it is a very deep model with well, structured convolutional layers.

Broken Soybean	112	4	4	21	19	0
Immature Soybean	2	139	16	9	2	0
Intact Soybean	8	27	124	17	4	0
Skin-damaged Soybean	13	4	12	139	1	0
Spotted Soybean	18	7	8	10	125	0
Healthy Soybean	0	0	0	0	0	179

Fig. 13. Confusion Matrix of ResNet50 Model

TABLE I
ACCURACY TABLE

Model	Accuracy
CNN	74%
VGG16	85%
MobileNetV2	80%
ResNet50	80%

Tables II, III, and IV give the class, wise precision, recall, and F1, score results, respectively, for the four models we've implemented in classifying six different categories of soybean seeds. These performance metrics help us understand the strengths of the models in recognizing particular seed classes.

First, from the Table II, we notice that the VGG16 model yields higher precision for a majority of the defective seed categories, especially for skin, damaged and intact soybean seeds. This suggests that VGG16 model tends to produce less false positives than other models. The CNN has lower precision in the case of broken and skin, damaged seeds which may indicate that it finds it hard to differentiate those two types of visually similar defects. All the models have achieved 100% precision for healthy soybean seeds which is indicative of a good visual separation of the class.

Table III (Recall) presents the extent to which the models can correctly identify all the actual samples of each class. VGG16 and ResNet50 hold up their recall well in most categories, especially in the case of immature and skin, damaged soybean seeds. The CNN model has a lower recall for spotted and intact seeds which means it is missing some detections.

TABLE II
PRECISION TABLE

Seed Category	Model			
	CNN	VGG16	MobileNetV2	ResNet50
Broken Soybean	55%	79%	78%	73%
Immature Soybean	78%	77%	80%	77%
Intact Soybean	73%	88%	71%	76%
Skin-damaged Soybean	64%	94%	73%	71%
Spotted Soybean	77%	77%	77%	77%
Healthy Soybean	100%	100%	100%	100%

Once more, all the models have scored perfect recall for healthy soybean seeds which was expected. Table IV (F1, score) is a performance metric that combines precision and recall to give an overall measure of performance. VGG16 produces the highest F1, scores in most defect categories, thus confirming its general robustness. MobileNetV2 and ResNet50 demonstrate performance that is both competitive and consistent, whereas the CNN model is only satisfactory and produces lower scores particularly for the complex defect classes. Altogether, these tables suggest that VGG16 gives the classes, wise most dependable performance of any of the models.

TABLE III
RECALL TABLE

Seed Category	Model			
	CNN	VGG16	MobileNetV2	ResNet50
Broken Soybean	78%	81%	66%	75%
Immature Soybean	62%	81%	78%	83%
Intact Soybean	64%	78%	76%	69%
Skin-damaged Soybean	73%	80%	78%	82%
Spotted Soybean	63%	92%	79%	73%
Healthy Soybean	100%	100%	100%	100%

TABLE IV
F1-SCORE TABLE

Seed Category	Model			
	CNN	VGG16	MobileNetV2	ResNet50
Broken Soybean	65%	80%	71%	74%
Immature Soybean	70%	79%	79%	80%
Intact Soybean	68%	82%	73%	72%
Skin-damaged Soybean	68%	87%	75%	76%
Spotted Soybean	69%	84%	78%	80%
Healthy Soybean	100%	100%	100%	100%

V. CONCLUSION

This paper presents a comprehensive deep learning-based framework for soybean seed quality classification into six categories. The dataset was created by merging manually captured images of healthy soybean seeds with defective seed images collected from publicly available Kaggle repositories. The overall methodology follows a structured pipeline that includes data preprocessing and standardization, data augmentation, model training, and systematic performance evaluation.

Results clearly demonstrate the effectiveness of deep learning model in distinguishing healthy and defective soybean seeds. VGG16 achieved the highest classification accuracy of 85%, its strong feature extraction capability. MobileNetV2 and ResNet50 both are give same accuracy .The CNN model give lower accuracy. Overall, the findings validate deep learning as a reliable solution for automated soybean seeds quality inspection, with strong potential to reduce manual effort and improve consistency in agricultural quality control system.

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